

# **Lab 3 Work**

## **UML Diagram Generation with AI**

Course: Automation of Scientific Research

Name: Bilal Zai Mohammad Wasim

Date: 06/01/2026

**Aim:**

Using a library management system as the case study, compare the quality of UML diagrams produced by various AI models and prompt levels. to assess the accuracy of the PlantUML code and the degree to which the diagrams satisfy the specifications.

**Models Used**

- AI models: ChatGPT (GPT-5.2), DeepSeek (3.2)

**System under study**

Library management system

Actors (example): Reader, Library, Admin.

**4. Use Case Diagram – Results****4.1 Prompt**

Prompt Text:

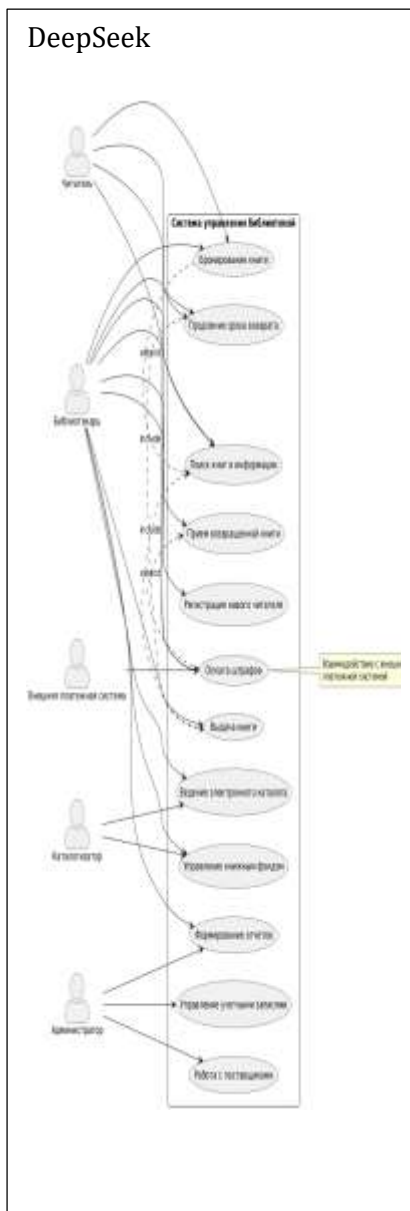
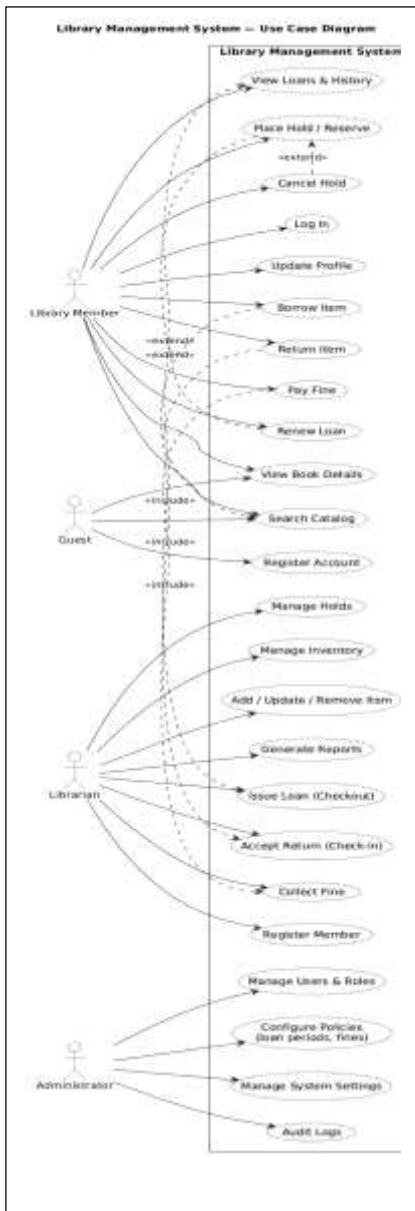
[Generate PlantUML code for a library management system use case diagram. Identify the actors and list the main use cases.]

| Model              | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Render (OK/Error) |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT (GPT-5.2), | @startuml<br>title Library Management System — Use Case Diagram<br>left to right direction<br>skinparam packageStyle rectangle<br><br>' Actors<br>actor "Library Member" as Member<br>actor "Librarian" as Librarian<br>actor "Administrator" as Admin<br>actor "Guest" as Guest<br><br>rectangle "Library Management System" {<br><br>' Guest / Public<br>usecase "Search Catalog" as UC_Search<br>usecase "View Book Details" as UC_ViewDetails<br>usecase "Register Account" as UC_Register<br><br>' Member<br>usecase "Log In" as UC_Login<br>usecase "Update Profile" as UC_Profile<br>usecase "Borrow Item" as UC_Borrow<br>usecase "Return Item" as UC_Return<br>usecase "Renew Loan" as UC_Renew<br>} | OK                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>usecase "Place Hold / Reserve" as UC_Reserve<br/> usecase "Cancel Hold" as UC_CancelHold<br/> usecase "View Loans &amp; History" as UC_History<br/> usecase "Pay Fine" as UC_PayFine</p> <p>' Librarian<br/> usecase "Issue Loan (Checkout)" as UC_Checkout<br/> usecase "Accept Return (Check-in)" as UC_Checkin<br/> usecase "Register Member" as UC_RegMember<br/> usecase "Manage Holds" as UC_ManageHolds<br/> usecase "Collect Fine" as UC_CollectFine<br/> usecase "Add / Update / Remove Item" as UC_ManageItems<br/> usecase "Manage Inventory" as UC_Inventory<br/> usecase "Generate Reports" as UC_Reports</p> <p>' Admin<br/> usecase "Manage Users &amp; Roles" as UC_Roles<br/> usecase "Configure Policies\n(loan periods, fines)" as UC_Policies<br/> usecase "Manage System Settings" as UC_Settings<br/> usecase "Audit Logs" as UC_Audit<br/> }</p> <p>' Associations<br/> Guest --&gt; UC_Search<br/> Guest --&gt; UC_ViewDetails<br/> Guest --&gt; UC_Register</p> <p>Member --&gt; UC_Login<br/> Member --&gt; UC_Search<br/> Member --&gt; UC_ViewDetails<br/> Member --&gt; UC_Profile<br/> Member --&gt; UC_Borrow<br/> Member --&gt; UC_Return<br/> Member --&gt; UC_Renew<br/> Member --&gt; UC_Reserve<br/> Member --&gt; UC_CancelHold<br/> Member --&gt; UC_History<br/> Member --&gt; UC_PayFine</p> <p>Librarian --&gt; UC_Checkout<br/> Librarian --&gt; UC_Checkin<br/> Librarian --&gt; UC_RegMember<br/> Librarian --&gt; UC_ManageHolds<br/> Librarian --&gt; UC_CollectFine<br/> Librarian --&gt; UC_ManageItems<br/> Librarian --&gt; UC_Inventory<br/> Librarian --&gt; UC_Reports</p> <p>Admin --&gt; UC_Roles<br/> Admin --&gt; UC_Policies<br/> Admin --&gt; UC_Settings<br/> Admin --&gt; UC_Audit</p> <p>' Includes / Extends (typical relationships)<br/> UC_Borrow .&gt; UC_Checkout : &lt;&lt;include&gt;&gt;<br/> UC_Return .&gt; UC_Checkin : &lt;&lt;include&gt;&gt;<br/> UC_PayFine .&gt; UC_CollectFine : &lt;&lt;include&gt;&gt;</p> <p>UC_Renew .&gt; UC_History : &lt;&lt;extend&gt;&gt;<br/> UC_Reserve .&gt; UC_Search : &lt;&lt;extend&gt;&gt;</p> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|                | UC_CancelHold .> UC_Reserve : <<extend>><br><br>@enduml                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |
| DeepSeek (3.2) | @startuml<br>left to right direction<br>skinparam actorStyle awesome<br><br>' Определение действующих лиц (актеров)<br>actor Библиотекарь as librarian<br>actor Читатель as reader<br>actor Администратор as admin<br>actor "Каталогизатор" as cataloger<br>actor "Внешняя платежная система" as payment<br><br>' Определение границ системы<br>rectangle "Система управления библиотекой" {<br><br>' Основные варианты использования, сгруппированные по актерам<br>(Поиск книг и информации) as UC1<br>(Регистрация нового читателя) as UC2<br>(Выдача книги) as UC3<br>(Прием возвращенной книги) as UC4<br>(Продление срока возврата) as UC5<br>(Формирование отчетов) as UC6<br>(Управление книжным фондом) as UC7<br>(Бронирование книги) as UC8<br>(Оплата штрафов) as UC9<br>(Управление учетными записями) as UC10<br>(Ведение электронного каталога) as UC11<br>(Работа с поставщиками) as UC12<br>}<br><br>' Связи актеров с вариантами использования<br><br>' Читатель<br>reader --> UC1<br>reader --> UC5<br>reader --> UC8<br>reader --> UC9<br><br>' Библиотекарь (основной пользователь)<br>librarian --> UC1<br>librarian --> UC2<br>librarian --> UC3<br>librarian --> UC4<br>librarian --> UC5<br>librarian --> UC9<br>librarian --> UC6<br>librarian --> UC7<br>librarian --> UC8<br>librarian --> UC11<br><br>' Каталогизатор<br>cataloger --> UC7<br>cataloger --> UC11<br><br>' Администратор<br>admin --> UC6<br>admin --> UC10 | ОК |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>admin --&gt; UC12</p> <p>' Внешняя система<br/>payment --&gt; UC9</p> <p>' Связи между вариантами использования (include/extend)</p> <p>' Базовые сценарии<br/>UC3 .&gt; UC1 : include<br/>UC8 .&gt; UC1 : include</p> <p>' Расширения (дополнительное поведение)<br/>UC9 .&gt; UC4 : extend<br/>UC5 .&gt; UC3 : extend</p> <p>' Примечание<br/>note right of UC9<br/>Взаимодействие с внешней<br/>платежной системой<br/>end note</p> <p>@enduml</p> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



The diagrams were correctly drawn, and both ChatGPT and DeepSeek generated valid PlantUML code (OK). Roles and use cases, including include/extend relations, are clearly separated in ChatGPT's more thorough diagram. Although DeepSeek's output is more condensed and simulates an external payment system, its applications are broader. For a comprehensive LMS use-case view, ChatGPT is superior overall.

## 4.2 Prompt

Prompt Text:

Сгенерируй PlantUML код для диаграммы вариантов использования системы управления библиотекой. Используй акторов: Читатель (Reader), Библиотекарь (Librarian), Администратор (Admin). Для каждого актора укажи 5–8 основных вариантов использования (use cases). Сделай границу системы (system boundary) "Library Management System". Дай полный код от @startuml до @enduml

| Model         | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Render (OK/Error) |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT (5.2) | <pre> @startuml title Library Management System - Use Case Diagram left to right direction skinparam packageStyle rectangle  actor "Читатель (Reader)" as Reader actor "Библиотекарь (Librarian)" as Librarian actor "Администратор (Admin)" as Admin  rectangle "Library Management System" {      ' Reader use cases (5–8)     usecase "Search Catalog" as UC_Search     usecase "View Item Details" as UC_ViewDetails     usecase "Borrow Item" as UC_Borrow     usecase "Return Item" as UC_Return     usecase "Renew Loan" as UC_Renew     usecase "Place Hold / Reserve" as UC_Reserve     usecase "View Loans &amp; History" as UC_History     usecase "Pay Fine" as UC_PayFine      ' Librarian use cases (5–8)     usecase "Register Reader" as UC_RegReader     usecase "Issue Loan (Checkout)" as UC_Checkout     usecase "Accept Return (Check-in)" as UC_Checkin     usecase "Manage Holds" as UC_ManageHolds     usecase "Collect Fine" as UC_CollectFine </pre> | OK                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> usecase "Add / Update / Remove Item" as UC_ManageItems usecase "Inventory Control" as UC_Inventory usecase "Generate Operational Reports" as UC_OpReports  ' Admin use cases (5-8) usecase "Manage Users &amp; Roles" as UC_UsersRoles usecase "Configure Loan Policies" as UC_Policies usecase "Configure Fine Rules" as UC_Fines usecase "Manage System Settings" as UC_Settings usecase "Audit Logs" as UC_Audit usecase "Backup / Restore Data" as UC_Backup usecase "Manage Integrations" as UC_Integrations usecase "Generate Administrative Reports" as UC_AdminReports }  ' Associations Reader --&gt; UC_Search Reader --&gt; UC_ViewDetails Reader --&gt; UC_Borrow Reader --&gt; UC_Return Reader --&gt; UC_Renew Reader --&gt; UC_Reserve Reader --&gt; UC_History Reader --&gt; UC_PayFine  Librarian --&gt; UC_RegReader Librarian --&gt; UC_Checkout Librarian --&gt; UC_Checkin Librarian --&gt; UC_ManageHolds Librarian --&gt; UC_CollectFine Librarian --&gt; UC_ManageItems Librarian --&gt; UC_Inventory Librarian --&gt; UC_OpReports  Admin --&gt; UC_UsersRoles Admin --&gt; UC_Policies Admin --&gt; UC_Fines Admin --&gt; UC_Settings Admin --&gt; UC_Audit Admin --&gt; UC_Backup Admin --&gt; UC_Integrations Admin --&gt; UC_AdminReports  ' Typical include relationships UC_Borrow .&gt; UC_Checkout : &lt;&lt;include&gt;&gt; UC_Return .&gt; UC_Checkin : &lt;&lt;include&gt;&gt; UC_PayFine .&gt; UC_CollectFine : &lt;&lt;include&gt;&gt; </pre> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

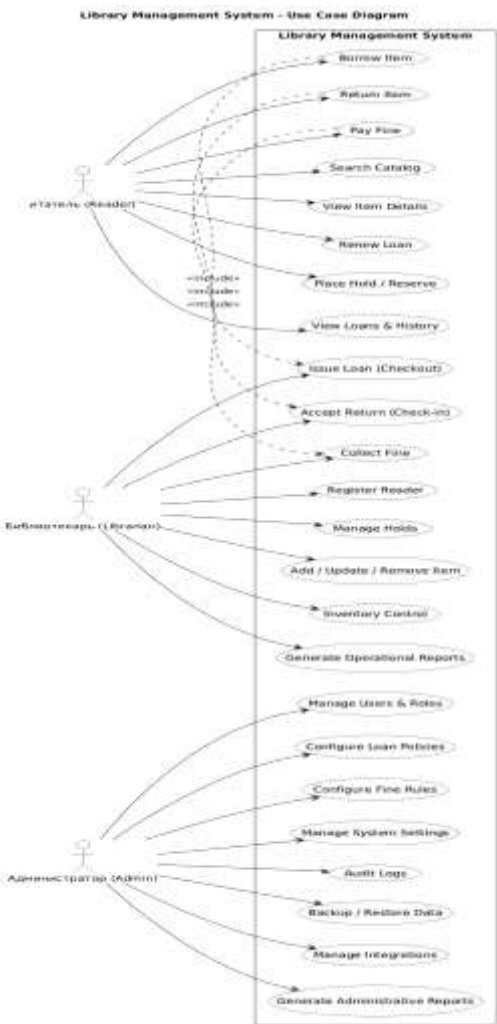
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                | @enduml                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| DeepSeek(3.2 ) | <pre> @startuml left to right direction skinparam packageStyle rectangle title Library Management System - Use Case Diagram  ' Define system boundary rectangle "Library Management System" { ' Define use cases for all actors within system boundary  (Search Catalog) as UC1 (Borrow Book) as UC2 (Return Book) as UC3 (Renew Loan) as UC4 (View Loan History) as UC5 (Place Hold) as UC6 (Pay Fine) as UC7 (Update Profile) as UC8  (Process Checkout) as UC9 (Process Return) as UC10 (Manage Holds) as UC11 (Manage Fines) as UC12 (Add New Book) as UC13 (Update Book Status) as UC14 (Generate Reports) as UC15 (Register New Member) as UC16  (Manage User Accounts) as UC17 (Manage Staff Accounts) as UC18 (Configure System Settings) as UC19 (Backup System Data) as UC20 (View System Logs) as UC21 (Manage Library Inventory) as UC22 (Set Loan Policies) as UC23 (Manage Library Branches) as UC24 }  ' Define actors actor "Reader" as reader actor "Librarian" as librarian actor "Admin" as admin  ' Reader connections reader --&gt; UC1 : &lt;&lt;extend&gt;&gt;\nSearch by title/author reader --&gt; UC2 : &lt;&lt;include&gt;&gt;\nRequires valid membership reader --&gt; UC3 reader --&gt; UC4 : &lt;&lt;extend&gt;&gt;\nIf no holds reader --&gt; UC5 </pre> | error |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>reader --&gt; UC6 : &lt;&lt;extend&gt;&gt;\nWhen book unavailable</p> <p>reader --&gt; UC7 : &lt;&lt;extend&gt;&gt;\nFor overdue items</p> <p>reader --&gt; UC8</p> <p>' Librarian connections</p> <p>librarian --&gt; UC9 : &lt;&lt;include&gt;&gt;\nRequires UC1</p> <p>librarian --&gt; UC10</p> <p>librarian --&gt; UC11</p> <p>librarian --&gt; UC12</p> <p>librarian --&gt; UC13</p> <p>librarian --&gt; UC14 : &lt;&lt;extend&gt;&gt;\nMark as lost/damaged</p> <p>librarian --&gt; UC15</p> <p>librarian --&gt; UC16</p> <p>' Admin connections</p> <p>admin --&gt; UC17</p> <p>admin --&gt; UC18</p> <p>admin --&gt; UC19</p> <p>admin --&gt; UC20</p> <p>admin --&gt; UC21</p> <p>admin --&gt; UC22</p> <p>admin --&gt; UC23</p> <p>admin --&gt; UC24</p> <p>' Define relationships between use cases</p> <p>UC2 .&gt; UC1 : &lt;&lt;include&gt;&gt;</p> <p>UC9 .&gt; UC1 : &lt;&lt;include&gt;&gt;</p> <p>UC7 .&gt; UC10 : &lt;&lt;extend&gt;&gt;</p> <p>UC6 .&gt; UC1 : &lt;&lt;include&gt;&gt;</p> <p>' Group use cases visually</p> <p>together {</p> <p>    UC1; UC2; UC3; UC4; UC5; UC6; UC7; UC8</p> <p>}</p> <p>together {</p> <p>    UC9; UC10; UC11; UC12; UC13; UC14; UC15; UC16</p> <p>}</p> <p>together {</p> <p>    UC17; UC18; UC19; UC20; UC21; UC22; UC23; UC24</p> <p>}</p> <p>' Add notes for clarification</p> <p>note top of UC2</p> <p>    Reader can borrow up to</p> <p>    10 items for 21 days</p> <p>end note</p> |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                         |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>note right of UC12<br/>Librarian can waive or adjust fines with reason<br/>end note</p> <p>note left of UC19<br/>Admin can configure loan periods, fine rates, and system parameters<br/>end note</p> <p>@enduml</p> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

ChatGPT



DeepSeek

```
PlantUML 1.2025.11beta10
[From string (line 81) ]

@startuml
left to right direction
skinparam packageStyle rectangle
title Library Management System - Use Case Diagram

...
... ( skipping 48 lines )
...
librarian --> UC14 : <<extend>>\nMark as lost/damaged
librarian --> UC15
librarian --> UC16

admin --> UC17
admin --> UC18
admin --> UC19
admin --> UC20
admin --> UC21
admin --> UC22
admin --> UC23
admin --> UC24

UC2 .-> UC1 : <<include>>
UC9 .-> UC1 : <<include>>
UC7 .-> UC10 : <<extend>>
UC6 .-> UC1 : <<include>>

together {
    UC1; UC2; UC3; UC4; UC5; UC6; UC7; UC8
}
Syntax Error? (Assumed diagram type: component)
```

The use case diagram was successfully drawn (OK), and ChatGPT generated valid PlantUML code. A syntax error produced by jointly { ... } prevented DeepSeek's initial output from rendering, and PlantUML assumed a different diagram type. The diagram can be produced once that portion has been deleted or rewritten. On the first try, ChatGPT was generally more dependable.

### 4.3 prompt:

#### Prompt text:

Create a journal-quality use case diagram (PlantUML) for a library management system. Include the following actors: Reader, Librarian, and Administrator. Use cases should cover: catalog searching, lending/returning/renewing books, reservations, fines payments, inventory management, user/role management, and reporting. Add appropriate <> and <> relationships where logical. Display the complete PlantUML code from @startuml to @enduml. No bulleted lists, no additional explanations—just the code

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Render (OK/Error) |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | <pre> @startuml title Library Management System - Use Case Diagram left to right direction skinparam packageStyle rectangle skinparam shadowing false  actor "Reader" as Reader actor "Librarian" as Librarian actor "Administrator" as Administrator  rectangle "Library Management System" as LMS {      usecase "Search Catalog" as UC_Search     usecase "View Book Details" as UC_ViewDetails      usecase "Borrow Book" as UC_Borrow     usecase "Return Book" as UC_Return     usecase "Renew Loan" as UC_Renew     usecase "Reserve Book" as UC_Reserve      usecase "Pay Fine" as UC_PayFine     usecase "Calculate Fine" as UC_CalcFine     usecase "Record Payment" as UC_RecordPayment      usecase "Manage Inventory" as UC_ManageInventory     usecase "Add/Update/Remove Book" as UC_BookCRUD     usecase "Update Book Status\n(available/loaned/lost)" as UC_BookStatus      usecase "Manage Users &amp; Roles" as UC_UserRoles     usecase "Manage Reader Accounts" as UC_ReaderAccounts     usecase "Authenticate User" as UC_Auth      usecase "Reporting" as UC_Reporting     usecase "Operational Reports" as UC_OpReports     usecase "Administrative Reports" as UC_AdminReports  } </pre> | OK                |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|              | <p>Reader --&gt; UC_Search<br/> Reader --&gt; UC_ViewDetails<br/> Reader --&gt; UC_Borrow<br/> Reader --&gt; UC_Return<br/> Reader --&gt; UC_Renew<br/> Reader --&gt; UC_Reserve<br/> Reader --&gt; UC_PayFine</p> <p>Librarian --&gt; UC_ManageInventory<br/> Librarian --&gt; UC_ReaderAccounts<br/> Librarian --&gt; UC_Reporting<br/> Librarian --&gt; UC_RecordPayment</p> <p>Administrator --&gt; UC_UserRoles<br/> Administrator --&gt; UC_Reporting<br/> Administrator --&gt; UC_AdminReports</p> <p>UC_Search ..&gt; UC_ViewDetails : &lt;&lt;include&gt;&gt;<br/> UC_Borrow ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_Return ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_Renew ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_Reserve ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_PayFine ..&gt; UC_CalcFine : &lt;&lt;include&gt;&gt;<br/> UC_PayFine ..&gt; UC_RecordPayment : &lt;&lt;include&gt;&gt;<br/> UC_ManageInventory ..&gt; UC_BookCRUD : &lt;&lt;include&gt;&gt;<br/> UC_ManageInventory ..&gt; UC_BookStatus : &lt;&lt;include&gt;&gt;<br/> UC_ReaderAccounts ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_UserRoles ..&gt; UC_Auth : &lt;&lt;include&gt;&gt;<br/> UC_Reporting ..&gt; UC_OpReports : &lt;&lt;include&gt;&gt;</p> <p>UC_Renew ..&gt; UC_Borrow : &lt;&lt;extend&gt;&gt;<br/> UC_Reserve ..&gt; UC_Search : &lt;&lt;extend&gt;&gt;<br/> UC_AdminReports ..&gt; UC_Reporting : &lt;&lt;extend&gt;&gt;</p> <p>@enduml</p> |    |
| DeepSeek 3.2 | <p>@startuml<br/> skinparam {<br/>   BackgroundColor #F8F9FA<br/>   ArrowColor #2C3E50<br/>   ActorBorderColor #2C3E50<br/>   UseCaseBorderColor #34495E<br/>   UseCaseBackgroundColor #ECF0F1<br/>   UseCaseFontColor #2C3E50<br/>   ActorFontColor #2C3E50<br/>   ActorBackgroundColor #E8F4FD<br/> }<br/> <br/> title Library Management System - Use Case Diagram<br/> <br/> left to right direction<br/> <br/> actor "Reader" &lt;&lt;Member&gt;&gt; as Reader<br/> actor "Librarian" &lt;&lt;Staff&gt;&gt; as Librarian<br/> actor "Administrator" &lt;&lt;Staff&gt;&gt; as Administrator<br/> <br/> rectangle "Library Management System" {<br/> <br/> ' Reader use cases<br/> (Search Catalog) as UC1<br/> (View Item Details) as UC2</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | OK |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>(Borrow Book) as UC3<br/> (Return Book) as UC4<br/> (Renew Loan) as UC5<br/> (Place Reservation) as UC6<br/> (Cancel Reservation) as UC7<br/> (View Loan History) as UC8<br/> (Pay Fine Online) as UC9<br/> (Update Personal Information) as UC10</p> <p>' Librarian use cases<br/> (Process Check-out) as UC11<br/> (Process Return) as UC12<br/> (Process Renewal) as UC13<br/> (Manage Reservations) as UC14<br/> (Calculate &amp; Apply Fines) as UC15<br/> (Accept Fine Payment) as UC16<br/> (Add New Inventory Item) as UC17<br/> (Update Item Status) as UC18<br/> (Remove Damaged/Lost Items) as UC19<br/> (Generate Circulation Reports) as UC20</p> <p>' Administrator use cases<br/> (Manage User Accounts) as UC21<br/> (Assign User Roles) as UC22<br/> (Define Fine Policies) as UC23<br/> (Set Loan Periods) as UC24<br/> (Configure System Parameters) as UC25<br/> (Backup/Restore System Data) as UC26<br/> (Generate Financial Reports) as UC27<br/> (Generate Inventory Reports) as UC28<br/> (Manage Library Branches) as UC29</p> <p>}</p> <p>' Reader connections<br/> Reader --&gt; UC1<br/> Reader --&gt; UC2<br/> Reader --&gt; UC3<br/> Reader --&gt; UC4<br/> Reader --&gt; UC5<br/> Reader --&gt; UC6<br/> Reader --&gt; UC7<br/> Reader --&gt; UC8<br/> Reader --&gt; UC9<br/> Reader --&gt; UC10</p> <p>' Librarian connections<br/> Librarian --&gt; UC11<br/> Librarian --&gt; UC12<br/> Librarian --&gt; UC13<br/> Librarian --&gt; UC14<br/> Librarian --&gt; UC15<br/> Librarian --&gt; UC16<br/> Librarian --&gt; UC17<br/> Librarian --&gt; UC18<br/> Librarian --&gt; UC19<br/> Librarian --&gt; UC20</p> <p>' Administrator connections<br/> Administrator --&gt; UC21<br/> Administrator --&gt; UC22</p> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | Administrator --> UC23<br>Administrator --> UC24<br>Administrator --> UC25<br>Administrator --> UC26<br>Administrator --> UC27<br>Administrator --> UC28<br>Administrator --> UC29<br><br>' Include relationships<br>UC3 .> UC1 : <<include>><br>UC11 .> UC1 : <<include>><br>UC11 .> UC3 : <<extend>><br>UC12 .> UC4 : <<extend>><br>UC13 .> UC5 : <<extend>><br>UC15 .> UC12 : <<include>><br>UC9 .> UC15 : <<extend>><br>UC16 .> UC9 : <<extend>><br>UC18 .> UC17 : <<extend>><br>UC6 .> UC1 : <<include>><br>UC7 .> UC6 : <<extend>><br>UC14 .> UC6 : <<include>><br>UC14 .> UC7 : <<extend>><br><br>' Extend relationships<br>UC5 .> UC3 : <<extend>><br>UC6 .> UC3 : <<extend>> \nwhen unavailable<br>UC15 .> UC4 : <<extend>> \non overdue items<br>UC18 .> UC19 : <<include>><br>UC21 .> UC22 : <<include>><br>UC24 .> UC23 : <<include>><br>UC28 .> UC17 : <<include>><br>UC28 .> UC18 : <<include>><br>UC28 .> UC19 : <<include>><br><br>note right of UC1<br>Search by:<br>- Title<br>- Author<br>- ISBN<br>- Subject<br>- Keyword<br>end note<br><br>note bottom of UC3<br>Maximum 10 items<br>Loan period: 21 days<br>end note<br><br>note left of UC15<br>Fine calculation based on:<br>- Overdue days<br>- Item type<br>- User category<br>end note<br><br>note top of UC21<br>Includes:<br>- Create account<br>- Deactivate account<br>- Update privileges<br>- Reset password |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



Both products are successfully presented (good), but ChatGPT provides a cleaner and more logically consistent use case model with clearer naming and more appropriate <<include>>/<<extend>> relationships (e.g., the shared "verify user"). DeepSec is more detailed and visually styled, but many of the relationships are questionable (e.g., "loan book includes search catalog" and mixed include/extend instructions), which may reduce the correctness of the UML. ChatGPT also maps well to the essential functions (search/loan/return/renewal/reserve/fines/inventory/user role/reporting) without unnecessary complexity. Overall, ChatGPT is a better choice for Prompt 3 due to its high clarity and UML consistency.

#### 4.4 Prompt:

Prompt text:

Create a journal quality use case diagram (PlantUML) for a library management system.

Actors: Reader, Librarian, Administrator.

Requirements:

Use cases should include: Search catalog, View item details, Borrow, Return, Renew, Cancel reservation, Pay fine.

Library should include: Check out/check in, Manage reservations, Manage inventory (add/update/remove), Generate reports.

Administrator should include: User and role management, Policy configuration (loan period/fine), Audit/Logs, Reporting.

Add realistic <<Include>> and <<Extend>> relationships (e.g., verification, fine calculation, early check-in).

Use system boundaries and keep the diagram readable (avoid excessive duplication).

Export only the complete PlantUML code from @startuml to @enduml.

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                       | Render (OK/Error) |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | <pre> @startuml title Library Management System - Use Case Diagram left to right direction skinparam packageStyle rectangle skinparam shadowing false skinparam usecase {   BackgroundColor White   BorderColor Black   ArrowColor Black }  actor "Reader" as Reader actor "Librarian" as Librarian </pre> | OK                |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>actor "Administrator" as Admin</p> <p>rectangle "Library Management System" as LMS {</p> <p>' Core catalog</p> <p>usecase "Search Catalog" as UC_Search</p> <p>usecase "View Item Details" as UC_ViewDetails</p> <p>' Reader circulation</p> <p>usecase "Borrow Item" as UC_Borrow</p> <p>usecase "Return Item" as UC_Return</p> <p>usecase "Renew Loan" as UC_Renew</p> <p>usecase "Cancel Reservation" as UC_CancelRes</p> <p>usecase "Pay Fine" as UC_PayFine</p> <p>' Library operations</p> <p>usecase "Check Out (Issue Loan)" as UC_CheckOut</p> <p>usecase "Check In (Accept Return)" as UC_CheckIn</p> <p>usecase "Manage Reservations" as UC_ManageRes</p> <p>usecase "Manage Inventory" as UC_Inventory</p> <p>usecase "Add/Update/Remove Item" as UC_ItemCRUD</p> <p>usecase "Generate Reports" as UC_Reports</p> <p>' Administration</p> <p>usecase "User &amp; Role Management" as UC_UserRoles</p> <p>usecase "Policy Configuration\n(loan period, fines)" as UC_Policy</p> <p>usecase "Audit / Logs" as UC_Audit</p> <p>usecase "Administrative Reporting" as UC_AdminReports</p> <p>' Shared/Supporting</p> <p>usecase "Verify Member Eligibility" as UC_Verify</p> <p>usecase "Validate Item Availability" as UC_Availability</p> <p>usecase "Calculate Fine" as UC_CalcFine</p> <p>usecase "Record Payment" as UC_RecordPay</p> <p>usecase "Update Item Status" as UC_UpdateStatus</p> <p>usecase "Notify Reader" as UC_Notify</p> <p>usecase "Early Check-in" as UC_EarlyCheckin</p> <p>}</p> <p>' Associations</p> <p>Reader --&gt; UC_Search</p> <p>Reader --&gt; UC_ViewDetails</p> <p>Reader --&gt; UC_Borrow</p> <p>Reader --&gt; UC_Return</p> <p>Reader --&gt; UC_Renew</p> <p>Reader --&gt; UC_CancelRes</p> <p>Reader --&gt; UC_PayFine</p> <p>Librarian --&gt; UC_CheckOut</p> <p>Librarian --&gt; UC_CheckIn</p> <p>Librarian --&gt; UC_ManageRes</p> <p>Librarian --&gt; UC_Inventory</p> <p>Librarian --&gt; UC_Reports</p> <p>Admin --&gt; UC_UserRoles</p> <p>Admin --&gt; UC_Policy</p> <p>Admin --&gt; UC_Audit</p> <p>Admin --&gt; UC_AdminReports</p> <p>' &lt;&lt;include&gt;&gt; (mandatory sub-functions)</p> <p>UC_Search ..&gt; UC_ViewDetails : &lt;&lt;include&gt;&gt;</p> |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

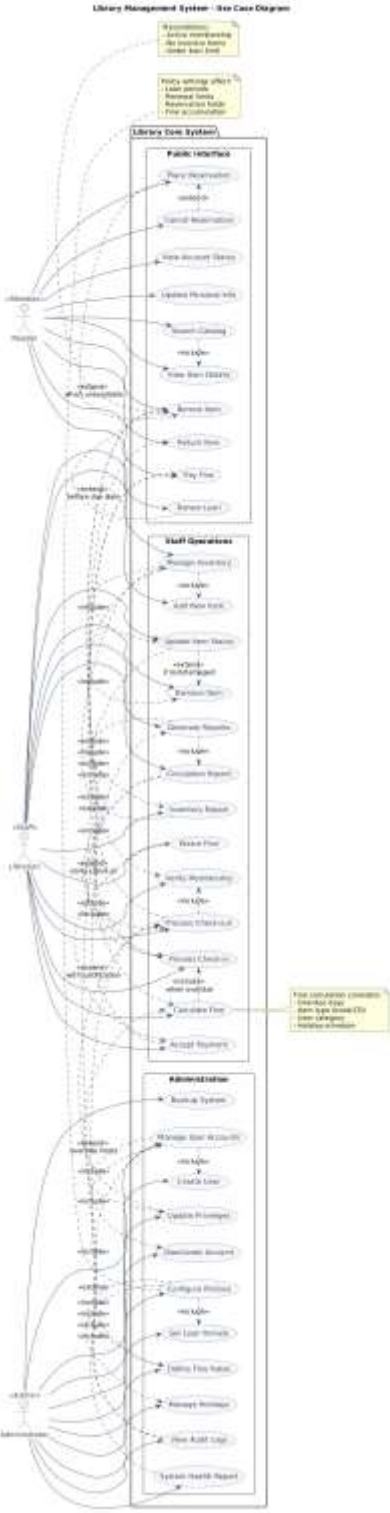
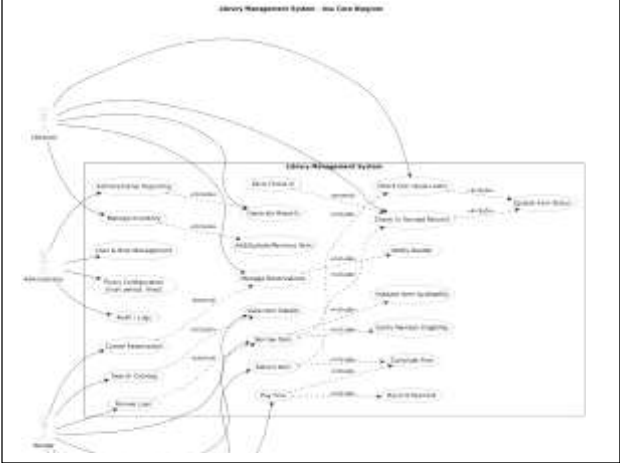
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|              | <pre> UC_Borrow ..&gt; UC_Verify : &lt;&lt;include&gt;&gt; UC_Borrow ..&gt; UC_Availability : &lt;&lt;include&gt;&gt; UC_Borrow ..&gt; UC_CheckOut : &lt;&lt;include&gt;&gt;  UC_Return ..&gt; UC_CheckIn : &lt;&lt;include&gt;&gt; UC_Return ..&gt; UC_CalcFine : &lt;&lt;include&gt;&gt;  UC_PayFine ..&gt; UC_CalcFine : &lt;&lt;include&gt;&gt; UC_PayFine ..&gt; UC_RecordPay : &lt;&lt;include&gt;&gt;  UC_CheckOut ..&gt; UC_UpdateStatus : &lt;&lt;include&gt;&gt; UC_CheckIn ..&gt; UC_UpdateStatus : &lt;&lt;include&gt;&gt;  UC_Inventory ..&gt; UC_ItemCRUD : &lt;&lt;include&gt;&gt; UC_ManageRes ..&gt; UC_Notify : &lt;&lt;include&gt;&gt; UC_AdminReports ..&gt; UC_Reports : &lt;&lt;include&gt;&gt;  ' &lt;&lt;extend&gt;&gt; (optional/conditional behavior) UC_Renew ..&gt; UC_Borrow : &lt;&lt;extend&gt;&gt; UC_CancelRes ..&gt; UC_ManageRes : &lt;&lt;extend&gt;&gt; UC_EarlyCheckin ..&gt; UC_CheckIn : &lt;&lt;extend&gt;&gt;  @enduml </pre>            |    |
| DeepSeek 3.2 | <pre> @startuml skinparam {   backgroundColor #FFFFFF   arrowColor #2C3E50   actorBorderColor #2C3E50   useCaseBorderColor #34495E   useCaseBackgroundColor #F8F9FA   useCaseFontColor #2C3E50   actorFontColor #2C3E50   actorBackgroundColor #EDF2F7 }  title Library Management System - Use Case Diagram  left to right direction  actor "Reader" &lt;&lt;Member&gt;&gt; as Reader actor "Librarian" &lt;&lt;Staff&gt;&gt; as Librarian actor "Administrator" &lt;&lt;Admin&gt;&gt; as Admin  package "Library Core System" {   rectangle "Public Interface" {     (Search Catalog) as UC1     (View Item Details) as UC2     (View Account Status) as UC3     (Update Personal Info) as UC4      (Borrow Item) as UC5     (Return Item) as UC6     (Renew Loan) as UC7     (Place Reservation) as UC8     (Cancel Reservation) as UC9     (Pay Fine) as UC10   }    rectangle "Staff Operations" {     (Verify Membership) as UC11     (Process Check-out) as UC12   } </pre> | OK |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | (Process Check-in) as UC13<br>(Calculate Fine) as UC14<br>(Accept Payment) as UC15<br>(Waive Fine) as UC16<br><br>(Manage Inventory) as UC17<br>(Add New Item) as UC18<br>(Update Item Status) as UC19<br>(Remove Item) as UC20<br><br>(Generate Reports) as UC21<br>(Circulation Report) as UC22<br>(Inventory Report) as UC23<br>}<br><br>rectangle "Administration" {<br>(Manage User Accounts) as UC24<br>(Create User) as UC25<br>(Update Privileges) as UC26<br>(Deactivate Account) as UC27<br><br>(Configure Policies) as UC28<br>(Set Loan Periods) as UC29<br>(Define Fine Rates) as UC30<br>(Manage Holidays) as UC31<br><br>(View Audit Logs) as UC32<br>(Backup System) as UC33<br>(System Health Report) as UC34<br>}<br>}<br><br>' Reader connections<br>Reader --> UC1<br>Reader --> UC2<br>Reader --> UC3<br>Reader --> UC4<br>Reader --> UC5<br>Reader --> UC6<br>Reader --> UC7<br>Reader --> UC8<br>Reader --> UC9<br>Reader --> UC10<br><br>' Librarian connections<br>Librarian --> UC11<br>Librarian --> UC12<br>Librarian --> UC13<br>Librarian --> UC14<br>Librarian --> UC15<br>Librarian --> UC16<br>Librarian --> UC17<br>Librarian --> UC18<br>Librarian --> UC19<br>Librarian --> UC20<br>Librarian --> UC21<br>Librarian --> UC22<br>Librarian --> UC23<br><br>' Administrator connections<br>Admin --> UC24 |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>Admin --&gt; UC25<br/> Admin --&gt; UC26<br/> Admin --&gt; UC27<br/> Admin --&gt; UC28<br/> Admin --&gt; UC29<br/> Admin --&gt; UC30<br/> Admin --&gt; UC31<br/> Admin --&gt; UC32<br/> Admin --&gt; UC33<br/> Admin --&gt; UC34</p> <p>' Include relationships<br/> UC1 .&gt; UC2 : &lt;&lt;include&gt;&gt;<br/> UC5 .&gt; UC11 : &lt;&lt;include&gt;&gt;<br/> UC12 .&gt; UC11 : &lt;&lt;include&gt;&gt;<br/> UC12 .&gt; UC5 : &lt;&lt;include&gt;&gt;<br/> UC13 .&gt; UC6 : &lt;&lt;include&gt;&gt;<br/> UC14 .&gt; UC13 : &lt;&lt;include&gt;&gt; \nwhen overdue<br/> UC15 .&gt; UC10 : &lt;&lt;include&gt;&gt;<br/> UC17 .&gt; UC18 : &lt;&lt;include&gt;&gt;<br/> UC17 .&gt; UC19 : &lt;&lt;include&gt;&gt;<br/> UC17 .&gt; UC20 : &lt;&lt;include&gt;&gt;<br/> UC21 .&gt; UC22 : &lt;&lt;include&gt;&gt;<br/> UC21 .&gt; UC23 : &lt;&lt;include&gt;&gt;<br/> UC24 .&gt; UC25 : &lt;&lt;include&gt;&gt;<br/> UC24 .&gt; UC26 : &lt;&lt;include&gt;&gt;<br/> UC24 .&gt; UC27 : &lt;&lt;include&gt;&gt;<br/> UC28 .&gt; UC29 : &lt;&lt;include&gt;&gt;<br/> UC28 .&gt; UC30 : &lt;&lt;include&gt;&gt;<br/> UC28 .&gt; UC31 : &lt;&lt;include&gt;&gt;</p> <p>' Extend relationships<br/> UC7 .&gt; UC5 : &lt;&lt;extend&gt;&gt; \nbefore due date<br/> UC8 .&gt; UC5 : &lt;&lt;extend&gt;&gt; \nwhen unavailable<br/> UC9 .&gt; UC8 : &lt;&lt;extend&gt;&gt;<br/> UC10 .&gt; UC14 : &lt;&lt;extend&gt;&gt;<br/> UC16 .&gt; UC15 : &lt;&lt;extend&gt;&gt; \nwith justification<br/> UC19 .&gt; UC13 : &lt;&lt;extend&gt;&gt; \nnearly check-in<br/> UC19 .&gt; UC20 : &lt;&lt;extend&gt;&gt; \nnif lost/damaged<br/> UC26 .&gt; UC12 : &lt;&lt;extend&gt;&gt; \nnoverride limits<br/> UC30 .&gt; UC14 : &lt;&lt;extend&gt;&gt;</p> <p>' Cross-package relationships<br/> UC22 .&gt; UC12 : &lt;&lt;include&gt;&gt;<br/> UC22 .&gt; UC13 : &lt;&lt;include&gt;&gt;<br/> UC23 .&gt; UC17 : &lt;&lt;include&gt;&gt;<br/> UC32 .&gt; UC12 : &lt;&lt;include&gt;&gt;<br/> UC32 .&gt; UC24 : &lt;&lt;include&gt;&gt;<br/> UC34 .&gt; UC21 : &lt;&lt;include&gt;&gt;</p> <p>note top of UC5<br/> Preconditions:<br/> - Active membership<br/> - No overdue items<br/> - Under loan limit<br/> end note</p> <p>note right of UC14<br/> Fine calculation considers:<br/> - Overdue days<br/> - Item type (book/CD)</p> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                         |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <ul style="list-style-type: none"><li>- User category</li><li>- Holiday schedule</li></ul> end note                                                                                                     |  |
|  | note bottom of UC28<br>Policy settings affect: <ul style="list-style-type: none"><li>- Loan periods</li><li>- Renewal limits</li><li>- Reservation holds</li><li>- Fine accumulation</li></ul> end note |  |
|  | @enduml                                                                                                                                                                                                 |  |

ChatGPT



Both ChatGPT and DeepSeek are well presented (well) and cover the essential actors and core functions. ChatGPT is cleaner and more readable, with well-chosen supporting use cases (capacity, availability, good accounting/payment) and constant <<include>>/<<extend>> links. DeepSeek is more comprehensive and well-structured into subsystems, but it is more complex and includes additional use cases that may be beyond the scope of the minimum. Overall, ChatGPT is better for short "journal quality" diagrams, while DeepSeek is useful when maximum detail is needed.

## 5. Class Diagram — Results

### 5.1 Prompt

Generate PlantUML code for a class diagram of a library management system.

Include key classes (at least 8-12), such as: library, catalog, book/item, user, reader, librarian, manager, loan, reservation, and fine/payment.

Show relationships between classes (unions and inheritance where appropriate).

Keep it simple (no features yet).

Send the complete PlantUML code only from @startuml to @enduml.

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Render (OK/Error) |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | <pre> @startuml title Library Management System - Class Diagram skinparam classAttributeIconSize 0 skinparam shadowing false  class Library class Catalog class Item class Book class User class Reader class Librarian class Manager class Loan class Reservation class Fine class Payment  ' Inheritance User &lt; -- Reader User &lt; -- Librarian User &lt; -- Manager  Item &lt; -- Book  ' Core ownership / containment Library "1" o-- "1" Catalog Library "1" o-- "0..*" Item Library "1" o-- "0..*" User </pre> | OK                |

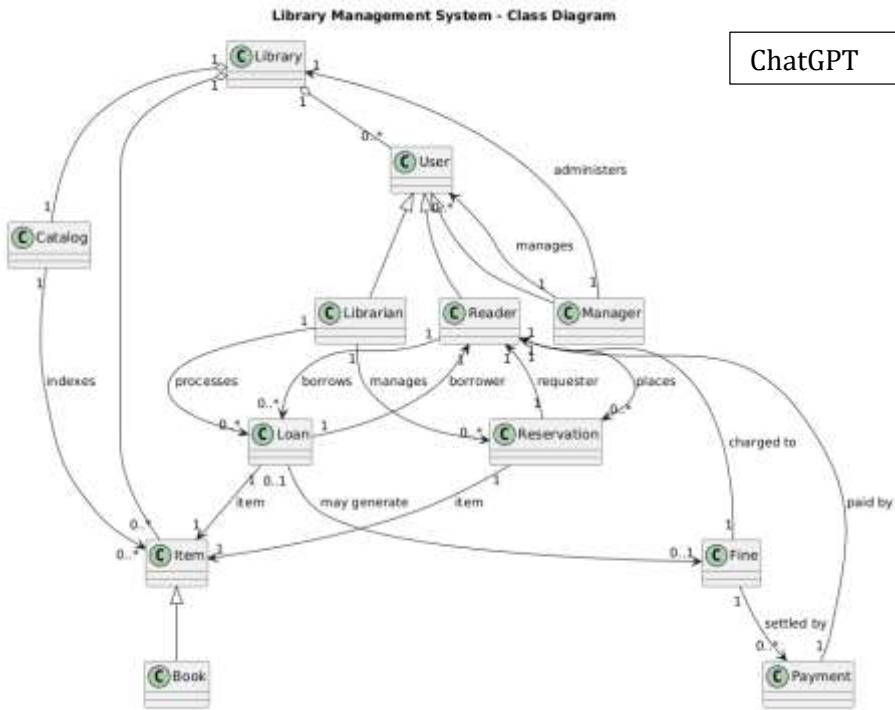
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|               | <p>' Catalog relationships<br/>Catalog "1" --&gt; "0..*" Item : indexes</p> <p>' Circulation<br/>Reader "1" --&gt; "0..*" Loan : borrows<br/>Loan "1" --&gt; "1" Item : item<br/>Loan "1" --&gt; "1" Reader : borrower</p> <p>Reader "1" --&gt; "0..*" Reservation : places<br/>Reservation "1" --&gt; "1" Item : item<br/>Reservation "1" --&gt; "1" Reader : requester</p> <p>' Fines and payments<br/>Loan "0..1" --&gt; "0..1" Fine : may generate<br/>Fine "1" --&gt; "1" Reader : charged to<br/>Fine "1" --&gt; "0..*" Payment : settled by<br/>Payment "1" --&gt; "1" Reader : paid by</p> <p>' Staff responsibilities<br/>Librarian "1" --&gt; "0..*" Loan : processes<br/>Librarian "1" --&gt; "0..*" Reservation : manages<br/>Manager "1" --&gt; "0..*" User : manages<br/>Manager "1" --&gt; "1" Library : administers</p> <p>@enduml</p> |    |
| DeepSeek(3.2) | <p>@startuml<br/>skinparam {<br/>  classBackgroundColor #F8F9FA<br/>  classBorderColor #2C3E50<br/>  classFontColor #2C3E50<br/>  arrowColor #2C3E50<br/>}<br/><br/>title Library Management System - Class Diagram</p> <pre> class Library { - name: String - address: String - phone: String + addBook(book: Book): void + removeBook(isbn: String): void + registerUser(user: User): void + searchCatalog(criteria: String): List&lt;Book&gt; }  class Catalog { - books: List&lt;Book&gt; + searchByTitle(title: String): List&lt;Book&gt; + searchByAuthor(author: String): List&lt;Book&gt; + searchByISBN(isbn: String): Book + addBook(book: Book): void + removeBook(book: Book): void }  class Book { - isbn: String - title: String - author: String - publicationYear: int - publisher: String - category: String </pre>                   | OK |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> - status: BookStatus + getStatus(): BookStatus + updateStatus(status: BookStatus): void }  enum BookStatus {     AVAILABLE     CHECKED_OUT     RESERVED     DAMAGED     LOST }  class Item {     - itemId: String     - acquisitionDate: Date     - location: String     + isAvailable(): boolean }  class User {     - userId: String     - name: String     - email: String     - phone: String     - registrationDate: Date     + getName(): String     + getEmail(): String     + updateProfile(): void }  class Reader {     - readerId: String     - membershipExpiry: Date     - maxBooksAllowed: int     + borrowBook(book: Book): Loan     + returnBook(loan: Loan): void     + renewLoan(loan: Loan): boolean }  class Librarian {     - employeeId: String     - hireDate: Date     + processCheckout(reader: Reader, book: Book): Loan     + processReturn(loan: Loan): void     + addBookToCatalog(book: Book): void     + generateReport(): Report }  class Manager {     - department: String     + manageInventory(): void     + generateFinancialReport(): FinancialReport     + manageStaff(): void }  class Loan {     - loanId: String     - borrowDate: Date     - dueDate: Date     - returnDate: Date </pre> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

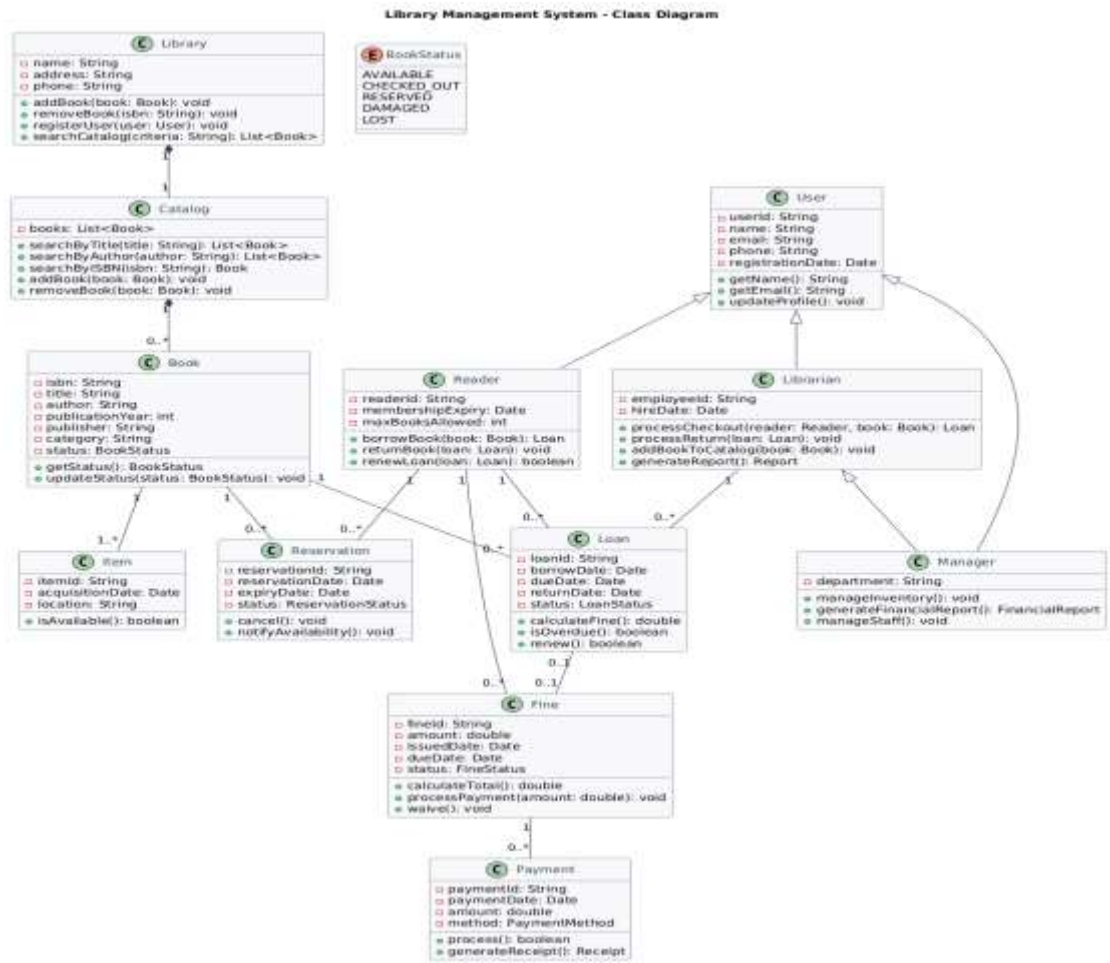


|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> - status: LoanStatus + calculateFine(): double + isOverdue(): boolean + renew(): boolean }  class Reservation { - reservationId: String - reservationDate: Date - expiryDate: Date - status: ReservationStatus + cancel(): void + notifyAvailability(): void }  class Fine { - fineId: String - amount: double - issuedDate: Date - dueDate: Date - status: FineStatus + calculateTotal(): double + processPayment(amount: double): void + waive(): void }  class Payment { - paymentId: String - paymentDate: Date - amount: double - method: PaymentMethod + process(): boolean + generateReceipt(): Receipt }  ' Inheritance relationships User &lt; -- Reader User &lt; -- Librarian User &lt; -- Manager Librarian &lt; -- Manager  ' Composition/Aggregation relationships Library "1" *-- "1" Catalog Catalog "1" *-- "0..*" Book Book "1" -- "1..*" Item  ' Associations Reader "1" -- "0..*" Loan Book "1" -- "0..*" Loan Librarian "1" -- "0..*" Loan Loan "0..1" -- "0..1" Fine Reader "1" -- "0..*" Reservation Book "1" -- "0..*" Reservation Fine "1" -- "0..*" Payment Reader "1" -- "0..*" Fine  @enduml </pre> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

ChatGPT



DeepSeek



Both diagrams are successfully rendered (good), but they differ in how well they meet the "simple/no attributes" requirement of Prompt 1. ChatGPT follows the prompt more closely by showing only key classes, inheritance, and parent relationships without attributes or methods. DeepSeq provides a more realistic and rich library model (for example, book-item separation and state concepts), but it goes beyond Prompt 1 by adding attributes, methods, and animations. Therefore, ChatGPT is the best choice for Prompt 1 compliance, while DeepSeek is better for a more detailed, implementation-based model.

## 5.2 Prompt:

Generate PlantUML code for a class diagram of a library management system.

Include key classes (library, catalog, item/book, user, reader, librarian, manager, loan, reservation, fine, payment).

Add 3-6 essential attributes to the main classes (e.g., ID, title, status, dates, size).

Keep relationships and inheritance clear, with multiplication where appropriate.

Export only the complete PlantUML code from @startuml to @enduml.

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                                                                                                                                                                | Render (OK/Error) |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | <pre> @startuml title Library Management System - Class Diagram skinparam classAttributeIconSize 0 skinparam shadowing false  ' ===== ' Core domain classes ' =====  class Library { +libraryId: String +name: String +address: String +phone: String }  class Catalog { +catalogId: String +name: String +lastUpdated: Date }  class Item { +itemId: String +barcode: String +title: String +status: String +location: String }  class Book { +isbn: String +author: String </pre> | OK                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>+publisher: String +publishedYear: Integer }  class User { +userId: String +fullName: String +email: String +phone: String +status: String }  class Reader { +membershipId: String +joinedDate: Date +maxLoans: Integer +currentLoans: Integer }  class Librarian { +employeeId: String +shift: String +workEmail: String }  class Manager { +employeeId: String +department: String +workEmail: String }  class Loan { +loanId: String +issueDate: Date +dueDate: Date +returnDate: Date +status: String }  class Reservation { +reservationId: String +requestDate: Date +expiryDate: Date +status: String }  class Fine { +fineId: String +amount: Decimal +reason: String +createdDate: Date +status: String }  class Payment { +paymentId: String +amount: Decimal +paymentDate: Date +method: String +referenceNo: String }</pre> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                      | <pre> ' ===== ' Inheritance ' ===== User &lt; -- Reader User &lt; -- Librarian User &lt; -- Manager  Item &lt; -- Book  ' ===== ' Aggregation / Composition ' ===== Library "1" o-- "1" Catalog : has Library "1" o-- "0..*" Item : holds Library "1" o-- "0..*" User : registers  Catalog "1" --&gt; "0..*" Item : indexes  ' ===== ' Circulation relationships ' ===== Reader "1" --&gt; "0..*" Loan : borrows Loan "1" --&gt; "1" Item : item Loan "1" --&gt; "1" Reader : borrower  Reader "1" --&gt; "0..*" Reservation : places Reservation "1" --&gt; "1" Item : item Reservation "1" --&gt; "1" Reader : requester  Loan "0..1" --&gt; "0..1" Fine : may generate Fine "1" --&gt; "1" Reader : charged to Fine "1" --&gt; "0..*" Payment : settled by Payment "1" --&gt; "1" Reader : paid by  ' ===== ' Staff responsibilities ' ===== Librarian "1" --&gt; "0..*" Loan : processes Librarian "1" --&gt; "0..*" Reservation : manages Manager "1" --&gt; "0..*" User : manages Manager "1" --&gt; "1" Library : administers  @enduml </pre> |           |
| <b>DeepSeek(3.2)</b> | <pre> @startuml skinparam {   classBackgroundColor #F8F9FA   classBorderColor #2C3E50   classFontColor #2C3E50   arrowColor #2C3E50 }  title Library Management System - Class Diagram  class Library {   - libraryId: String   - name: String   - address: String   - phone: String   - email: String   + addItem(item: Item): void </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>OK</b> |

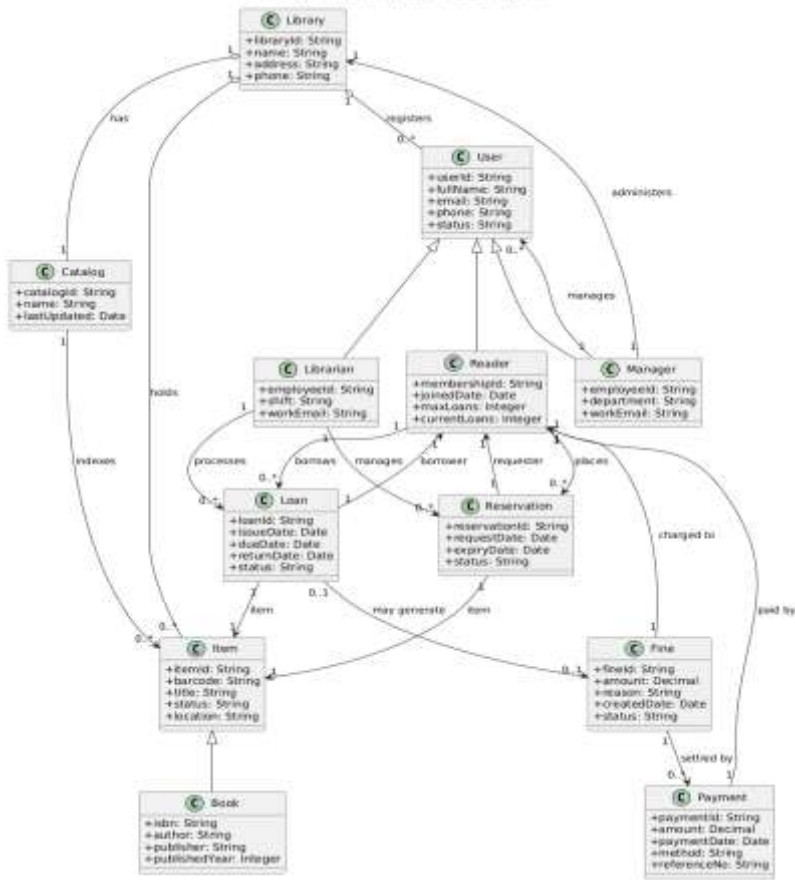
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> + registerUser(user: User): void }  class Catalog { - catalogId: String - lastUpdated: Date - totalItems: int + searchByTitle(title: String): List&lt;Item&gt; + searchByAuthor(author: String): List&lt;Item&gt; + searchByISBN(isbn: String): Item }  class Item { - itemId: String - isbn: String - title: String - author: String - publicationYear: int - status: ItemStatus + isAvailable(): boolean + updateStatus(status: ItemStatus): void }  class Book { - publisher: String - edition: String - pages: int - category: String }  class Magazine { - issueNumber: int - publicationDate: Date - frequency: String }  class DVD { - director: String - duration: int - rating: String }  class User { - userId: String - name: String - email: String - phone: String - registrationDate: Date - status: UserStatus + updateProfile(): void + getBorrowedItems(): List&lt;Loan&gt; }  class Reader { - membershipNumber: String - membershipExpiry: Date - maxLoans: int - totalFines: double + borrowItem(item: Item): Loan + returnItem(loan: Loan): void } </pre> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> class Librarian { - employeeId: String - hireDate: Date - department: String - accessLevel: int + processCheckout(reader: Reader, item: Item): Loan + processReturn(loan: Loan): void}  class Manager { - managerId: String - responsibilities: String[] + generateFinancialReport(): Report + manageStaff(): void }  class Loan { - loanId: String - borrowDate: Date - dueDate: Date - returnDate: Date - status: LoanStatus + calculateFine(): double + isOverdue(): boolean }  class Reservation { - reservationId: String - reservationDate: Date - expiryDate: Date - priority: int - status: ReservationStatus + cancel(): void + notifyUser(): void }  class Fine { - fineId: String - amount: double - issuedDate: Date - dueDate: Date - status: FineStatus + calculateTotal(): double + processPayment(): boolean }  class Payment { - paymentId: String - paymentDate: Date - amount: double - method: PaymentMethod - transactionId: String + process(): boolean }  ' Enum definitions enum ItemStatus { AVAILABLE CHECKED_OUT RESERVED DAMAGED </pre> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

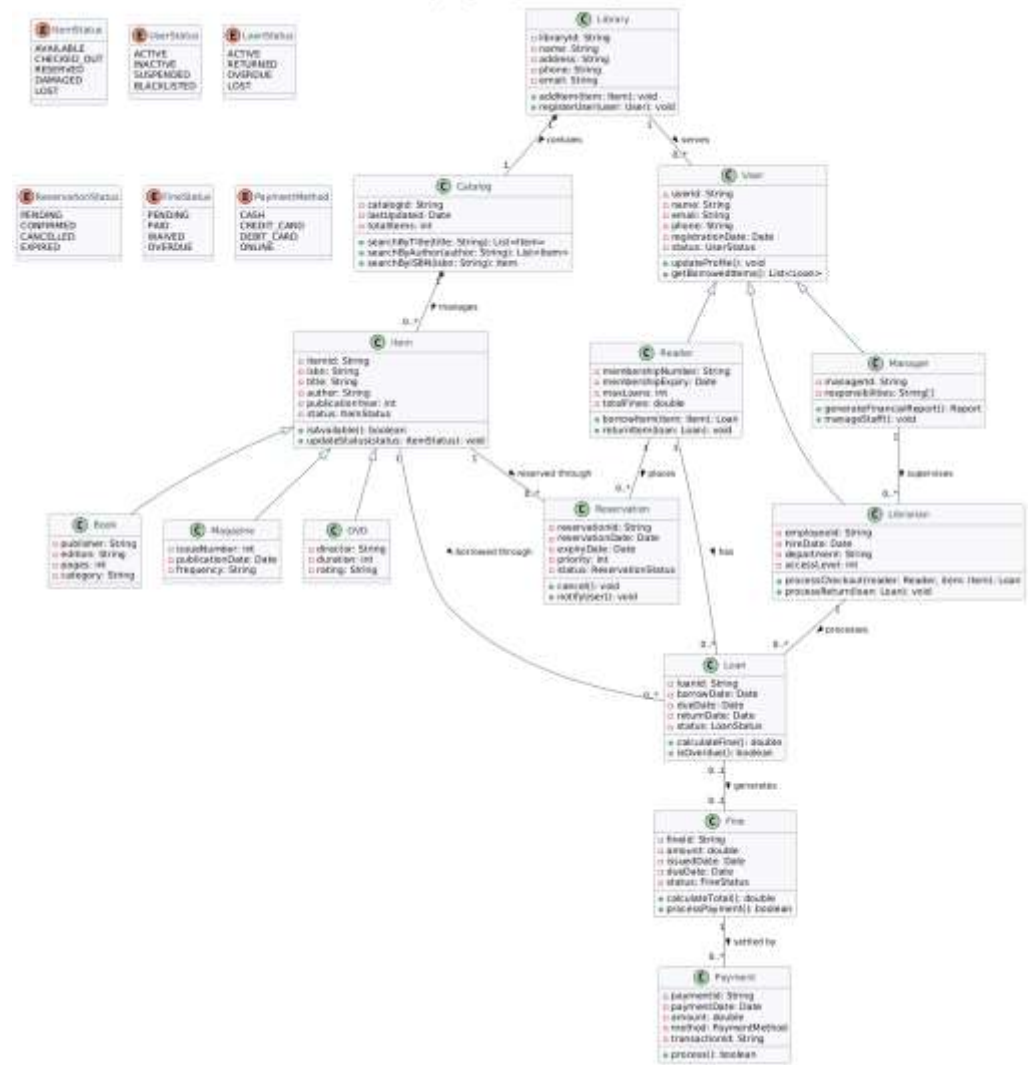
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>       LOST     }      enum UserStatus {       ACTIVE       INACTIVE       SUSPENDED       BLACKLISTED     }      enum LoanStatus {       ACTIVE       RETURNED       OVERDUE       LOST     }      enum ReservationStatus {       PENDING       CONFIRMED       CANCELLED       EXPIRED     }      enum FineStatus {       PENDING       PAID       WAIVED       OVERDUE     }      enum PaymentMethod {       CASH       CREDIT_CARD       DEBIT_CARD       ONLINE     }      ' Inheritance relationships     User &lt; -- Reader     User &lt; -- Librarian     User &lt; -- Manager     Item &lt; -- Book     Item &lt; -- Magazine     Item &lt; -- DVD      ' Composition relationships     Library "1" *-- "1" Catalog : contains &gt;     Catalog "1" *-- "0..*" Item : manages &gt;      ' Associations with multiplicities     Library "1" -- "0..*" User : serves &gt;     Reader "1" -- "0..*" Loan : has &gt;     Item "1" -- "0..*" Loan : borrowed through &gt;     Loan "0..1" -- "0..1" Fine : generates &gt;     Fine "1" -- "0..*" Payment : settled by &gt;     Reader "1" -- "0..*" Reservation : places &gt;     Item "1" -- "0..*" Reservation : reserved through &gt;     Librarian "1" -- "0..*" Loan : processes &gt;     Manager "1" -- "0..*" Librarian : supervises &gt;      @enduml </pre> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



Library Management System - Class Diagram



Library Management System - Class Diagram



The ChatGPT diagram is simple and well-suited to the stated requirements, focusing on core classes, clear inheritance, and essential relationships without unnecessary complexity.

The DeepSeek diagram is more detailed and feature-rich, but includes methods, animations, and additional classes that go beyond the basic class diagram.

In terms of readability and journal-style presentation, the ChatGPT version is clear and easy to interpret.

Overall, the ChatGPT model is more suitable for high-level academic or introductory system design.

5.3 Prompt:

Generate PlantUML code for a UML class diagram of a library management system.

Requirements:

Use inheritance: User as the base class, with Reader, Librarian, and Administrator as subclasses.

Model Book (library record) and Item (physical copy) appropriately, with clear relationships.

Use and label relationship types: Relationship, Collection, and Composition where appropriate.

Add multiples (e.g., 1, 0.., 1..).

Include at least one enum (e.g., ItemStatus or Loan Status).

Keep it readable and consistent with the library domain logic.

Only pass PlantUML code from @startuml to @enduml.

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                      | Render (OK/Error) |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | @startuml<br>title Library Management System - UML Class Diagram<br>skinparam classAttributeIconSize 0<br>skinparam shadowing false<br>left to right direction<br><br>' =====<br>' Enums<br>' =====<br><br>enum ItemStatus {<br>AVAILABLE<br>ON_LOAN<br>} | OK                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre>RESERVED LOST DAMAGED }  enum LoanStatus {   ACTIVE   RETURNED   OVERDUE }  ' ===== ' Users (Inheritance) ' ===== abstract class User {   +userId: String   +fullName: String   +email: String   +phone: String   +status: String }  class Reader {   +membershipId: String   +joinedDate: Date   +maxLoans: Integer }  class Librarian {   +employeeId: String   +workEmail: String   +shift: String }  class Administrator {   +employeeId: String   +workEmail: String   +roleLevel: Integer }  User &lt; -- Reader User &lt; -- Librarian User &lt; -- Administrator  ' ===== ' Bibliographic vs Physical ' ===== class Book {   +isbn: String   +title: String</pre> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> +author: String +publisher: String +publishedYear: Integer }  class Item { +itemId: String +barcode: String +status: ItemStatus +location: String +acquiredDate: Date }  ' ===== ' Circulation ' =====  class Loan { +loanId: String +issueDate: Date +dueDate: Date +returnDate: Date +status: LoanStatus }  class Reservation { +reservationId: String +requestDate: Date +expiryDate: Date +status: String }  class Fine { +fineId: String +amount: Decimal +createdDate: Date +status: String }  class Payment { +paymentId: String +amount: Decimal +paymentDate: Date +method: String +referenceNo: String }  class Library { +libraryId: String +name: String +address: String </pre> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|               | <pre> }  class Catalog { +catalogId: String +name: String +lastUpdated: Date }  ' ===== ' Labeled relationship types: ' Relationship, Collection, Composition ' =====  ' Composition (strong ownership) Library "1" *-- "1" Catalog : Composition Library "1" *-- "0..*" Item : Composition  ' Collection (managed grouping, weaker than composition) Catalog "1" o-- "0..*" Book : Collection Book "1" o-- "0..*" Item : Collection (copies)  ' Relationship (associations) Library "1" -- "0..*" User : Relationship (registered)  Reader "1" -- "0..*" Loan : Relationship (borrows) Loan "1" -- "1" Item : Relationship (item)  Reader "1" -- "0..*" Reservation : Relationship (places) Reservation "1" -- "1" Book : Relationship (for title)  Loan "0..1" -- "0..1" Fine : Relationship (may generate) Fine "1" -- "0..*" Payment : Relationship (settled by) Payment "1" -- "1" Reader : Relationship (paid by)  Librarian "1" -- "0..*" Loan : Relationship (processes) Administrator "1" -- "1" Library : Relationship (administers)  @enduml </pre> |    |
| DeepSeek(3.2) | <pre> @startuml skinparam { classBackgroundColor #F8F9FA classBorderColor #2C3E50 classFontColor #2C3E50 arrowColor #2C3E50 stereotypeFontColor #E74C3C } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | OK |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>title Library Management System - Class Diagram</p> <p>' Enums</p> <pre>enum ItemStatus {     AVAILABLE     CHECKED_OUT     RESERVED     DAMAGED     LOST     UNDER_REPAIR }</pre> <p>enum LoanStatus {</p> <pre>    ACTIVE     RETURNED     OVERDUE     LOST }</pre> <p>enum UserType {</p> <pre>    STUDENT     FACULTY     STAFF     GUEST }</pre> <p>' Base classes</p> <pre>class User {     - userId: String     - name: String     - email: String     - phone: String     - registrationDate: Date     - userType: UserType     - isActive: boolean     + authenticate(): boolean     + updateProfile(): void }</pre> <pre>class Book {     - isbn: String     - title: String     - author: String     - publisher: String     - publicationYear: int     - edition: String     - subject: String     + searchSimilar(): List&lt;Book&gt;     + getAvailableCopies(): List&lt;Item&gt; }</pre> <pre>class Item {     - itemId: String     - barcode: String     - acquisitionDate: Date     - location: String     - status: ItemStatus     - condition: String     + checkAvailability(): boolean     + updateStatus(status: ItemStatus): void }</pre> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> ' Subclasses class Reader { - readerId: String - membershipExpiry: Date - maxLoansAllowed: int - currentLoansCount: int + borrowItem(item: Item): Loan + returnItem(loan: Loan): void + renewLoan(loan: Loan): boolean + getBorrowingHistory(): List&lt;Loan&gt; }  class Librarian { - employeeId: String - hireDate: Date - department: String + processCheckout(reader: Reader, item: Item): Loan + processReturn(loan: Loan): void + addItemToCatalog(book: Book, item: Item): void + generateCirculationReport(): Report }  class Administrator { - adminId: String - accessLevel: int + manageUserAccounts(): void + configureSystemSettings(): void + generateFinancialReports(): Report + backupDatabase(): void }  ' Related classes class Loan { - loanId: String - checkoutDate: Date - dueDate: Date - returnDate: Date - status: LoanStatus + calculateFine(): double + isOverdue(): boolean + renew(): boolean }  class Reservation { - reservationId: String - reservationDate: Date - expiryDate: Date - priority: int + cancel(): void + notifyAvailable(): boolean }  class Fine { - fineId: String - amount: double - issuedDate: Date - dueDate: Date - reason: String + calculateTotal(): double + processPayment(amount: double): boolean </pre> |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

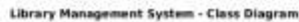
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> }  class Catalog { - catalogId: String - lastUpdated: Date + searchByTitle(title: String): List&lt;Book&gt; + searchByAuthor(author: String): List&lt;Book&gt; + searchBySubject(subject: String): List&lt;Book&gt; }  ' Inheritance relationships User &lt; -- Reader User &lt; -- Librarian User &lt; -- Administrator  ' Composition relationships Book "1" *-- "1..*" Item : has copies &gt; Catalog "1" *-- "0..*" Book : contains &gt;  ' Association relationships Reader "1" -- "0..*" Loan : "borrows" &gt; Item "1" -- "0..*" Loan : "loaned through" &gt; Librarian "1" -- "0..*" Loan : "processes" &gt; Loan "0..1" -- "0..1" Fine : "may incur" &gt; Reader "1" -- "0..*" Reservation : "places" &gt; Book "1" -- "0..*" Reservation : "reserved for" &gt; Item "0..1" -- "0..1" Reservation : "fulfilled by" &gt; Administrator "1" -- "0..*" Librarian : "manages" &gt;  ' Aggregation relationship Catalog "1" o-- "0..*" Item : "tracks" &gt;  ' Multiplicities Reader "1" --&gt; "0..5" Loan : "active loans" Item "1" --&gt; "0..1" Reader : "checked out to" Book "1" --&gt; "1..*" Item : "physical copies"  note "Item represents a physical copy\nof a Book (library record)" as N1 N1 .. Book N1 .. Item  note "Fine is generated when\nLoan becomes overdue" as N2 N2 .. Loan N2 .. Fine  @enduml </pre> |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



ChatGPT



DeepSeek



Both diagrams meet the requirements of Prompt 3 (inheritance from user, book-item modeling, anime, and multiplication) and render successfully (good). ChatGPT is cleaner and more "report friendly", but it uses relationship labels like "composition/collection" as text rather than really distinguishing semantics beyond standard UML arrows. DeepSeek implements relationship types consistently (composition book-item, collection/union) and adds useful domain constraints (e.g., active loan constraints), which make the model more realistic. Overall, DeepSeek is stronger for Prompt 3 due to its clear UML semantics and rich, consistent relationships.

#### 5.4 Prompt:

Produce a journal-quality UML class diagram (PlantUML) for a library management system.

Requirements:

Inheritance: User as base class; Reader, Librarian, Manager as subclasses.

Separate book (book record) and item (physical copy), with proper relationships.

Include classes: Library, Catalog, Loan, Reservation, Fine, and Payment.

Add key attributes with types (3-8 per class) and required operations (1-4 per class).

Use composition/collection/union as appropriate and add multiplication.

Include at least one enum for status (e.g., ItemStatus, Loan Status).

Keep it readable and consistent with domain logic.

Export only the complete PlantUML code from @startuml to @enduml.

| Model        | Plant UML code (@startuml...@enduml)                                                                                                                                                                                                                                                                                                    | Render (OK/Error) |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ChatGPT(5.2) | <pre> @startuml title Library Management System - UML Class Diagram left to right direction skinparam classAttributeIconSize 0 skinparam shadowing false skinparam packageStyle rectangle  ' ===== ' Enums ' ===== enum ItemStatus {     AVAILABLE     ON_LOAN     RESERVED     LOST     DAMAGED }  enum LoanStatus {     ACTIVE </pre> | OK                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> RETURNED OVERDUE LOST }  enum ReservationStatus {     PENDING     CONFIRMED     CANCELLED     EXPIRED }  ' ===== ' Users (Inheritance) ' =====  abstract class User {     +userId: String     +fullName: String     +email: String     +phone: String     +status: String     +authenticate(): boolean     +updateProfile(email: String, phone: String): void }  class Reader {     +membershipId: String     +joinedDate: Date     +maxLoans: Integer     +borrowItem(item: Item): Loan     +returnItem(loan: Loan): void     +payFine(fine: Fine, payment: Payment): void }  class Librarian {     +employeeId: String     +workEmail: String     +shift: String     +checkOut(reader: Reader, item: Item): Loan     +checkIn(loan: Loan): void     +manageInventory(item: Item): void }  class Manager {     +employeeId: String     +workEmail: String     +department: String     +configurePolicy(loanDays: Integer, fineRate: Decimal): void     +generateReport(type: String): String     +auditActivity(): void }  User &lt; -- Reader User &lt; -- Librarian User &lt; -- Manager  ' ===== ' Core library structure ' =====  class Library {     +libraryId: String </pre> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> +name: String +address: String +phone: String +openHours: String +registerUser(user: User): void +addItem(item: Item): void +removeItem(itemId: String): void }  class Catalog { +catalogId: String +name: String +lastUpdated: Date +searchByTitle(title: String): List&lt;Book&gt; +searchByAuthor(author: String): List&lt;Book&gt; +findByISBN(isbn: String): Book }  ' ===== ' Bibliographic vs Physical ' =====  class Book { +isbn: String +title: String +author: String +publisher: String +publishedYear: Integer +language: String +getAvailableCopies(): Integer }  class Item { +itemId: String +barcode: String +status: ItemStatus +location: String +acquiredDate: Date +markLost(): void +updateStatus(status: ItemStatus): void }  ' ===== ' Circulation / finance ' =====  class Loan { +loanId: String +issueDate: Date +dueDate: Date +returnDate: Date +status: LoanStatus +isOverdue(onDate: Date): boolean +close(returnDate: Date): void +calculateFine(onDate: Date): Decimal }  class Reservation { +reservationId: String +requestDate: Date +expiryDate: Date +status: ReservationStatus +confirm(): void </pre> |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|               | <pre> +cancel(): void +isExpired(onDate: Date): boolean }  class Fine { +fineId: String +amount: Decimal +reason: String +createdDate: Date +status: String +applyPayment(payment: Payment): void +getOutstandingAmount(): Decimal }  class Payment { +paymentId: String +amount: Decimal +paymentDate: Date +method: String +referenceNo: String +process(): boolean }  ' ===== ' Relationships (with types &amp; multiplicities) ' =====  ' Composition: Library strongly owns these Library "1" *-- "1" Catalog : composition Library "1" *-- "0..*" Item : composition  ' Collection: Catalog contains bibliographic records Catalog "1" o-- "0..*" Book : collection  ' Collection: A Book has multiple physical copies (Items) Book "1" o-- "0..*" Item : collection (copies)  ' Union/Association: Library registers users (membership) Library "1" -- "0..*" User : union/association (registered)  ' Loans Reader "1" -- "0..*" Loan : association (borrows) Loan "1" -- "1" Item : association (for copy) Loan "1" -- "1" Reader : association (borrower)  ' Reservations (for bibliographic record) Reader "1" -- "0..*" Reservation : association (places) Reservation "1" -- "1" Book : association (for title)  ' Fines and payments Loan "0..1" -- "0..1" Fine : association (may generate) Fine "1" -- "0..*" Payment : association (settled by) Payment "1" -- "1" Reader : association (paid by)  ' Staff responsibilities Librarian "1" -- "0..*" Loan : association (processes) Manager "1" -- "1" Library : association (administers)  @enduml </pre> |    |
| DeepSeek(3.2) | <pre> @startuml skinparam { </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OK |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> BackgroundColor #FFFFFF ClassBackgroundColor #F8F9FA ClassBorderColor #2C3E50 ClassFontColor #2C3E50 ArrowColor #2C3E50 StereotypeFontColor #8B4513 Shadowing false }  title Library Management System - Class Diagram  ' ===== ENUMS ===== enum ItemStatus {     AVAILABLE     CHECKED_OUT     RESERVED     DAMAGED     LOST     UNDER_REPAIR }  enum LoanStatus {     ACTIVE     RETURNED     OVERDUE     LOST }  enum UserRole {     STUDENT     FACULTY     STAFF     VIP }  enum PaymentStatus {     PENDING     COMPLETED     FAILED     REFUNDED }  ' ===== CORE CLASSES ===== class Library {     - libraryId: String     - name: String     - address: String     - phone: String     - email: String     - openingHours: String     + registerNewUser(user: User): boolean     + addBookRecord(book: Book): boolean     + generateMonthlyReport(): Report     + getSystemStatistics(): Statistics }  class Catalog {     - catalogId: String     - lastUpdated: Date     - totalTitles: int </pre> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> - totalCopies: int + searchByTitle(title: String): List&lt;Book&gt; + searchByAuthor(author: String): List&lt;Book&gt; + searchByISBN(isbn: String): Book + addNewBook(book: Book): boolean + updateBookInfo(isbn: String, updates: Map): boolean }  class Book { - isbn: String - title: String - author: String - publisher: String - publicationYear: int - edition: int - category: String - summary: String + getAvailableCopies(): List&lt;Item&gt; + getTotalCopies(): int + updateCatalogInfo(): boolean }  class Item { - itemId: String - barcode: String - acquisitionDate: Date - location: String - condition: String - status: ItemStatus + checkAvailability(): boolean + updateStatus(newStatus: ItemStatus): void + getBookRecord(): Book + calculateReplacementCost(): double }  ' ===== USER HIERARCHY ===== ===== class User { - userId: String - name: String - email: String - phone: String - registrationDate: Date - lastLogin: Date - isActive: boolean + authenticate(credentials: String): boolean + updateProfile(info: Map): boolean + changePassword(newPassword: String): boolean }  class Reader { - readerId: String - role: UserRole - membershipExpiry: Date - maxLoans: int - currentLoans: int - totalFines: double + borrowItem(item: Item): Loan + returnItem(loan: Loan): boolean + renewLoan(loan: Loan): boolean + placeReservation(book: Book): Reservation </pre> |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

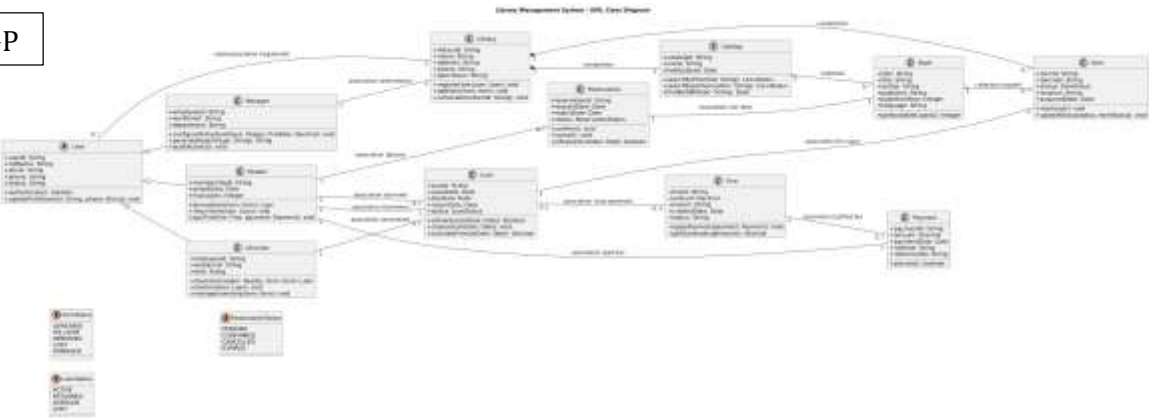
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> + getBorrowingHistory(): List&lt;Loan&gt; }  class Librarian { - employeeId: String - hireDate: Date - department: String - accessLevel: int + processCheckout(reader: Reader, item: Item): Loan + processReturn(loan: Loan): boolean + addPhysicalItem(book: Book, itemInfo: Map): Item + waiveFine(fine: Fine, reason: String): boolean + generateDailyReport(): Report }  class Manager { - managerId: String - responsibilities: List&lt;String&gt; - budgetAuthority: double + manageStaff(librarians: List&lt;Librarian&gt;): void + approveExpenses(amount: double): boolean + generateFinancialReport(): FinancialReport + setSystemParameters(params: Map): boolean }  ' ===== TRANSACTION CLASSES ===== class Loan { - loanId: String - checkoutDate: Date - dueDate: Date - returnDate: Date - status: LoanStatus - renewalsUsed: int + calculateFine(): double + isOverdue(): boolean + renew(): boolean + getDaysOverdue(): int }  class Reservation { - reservationId: String - reservationDate: Date - expiryDate: Date - priority: int - notificationSent: boolean + cancel(): boolean + notifyUser(): boolean + checkAvailability(): boolean + getQueuePosition(): int }  class Fine { - fineId: String - amount: double - issuedDate: Date - dueDate: Date - reason: String - status: PaymentStatus + calculateTotal(): double + applyDiscount(percentage: double): void </pre> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|



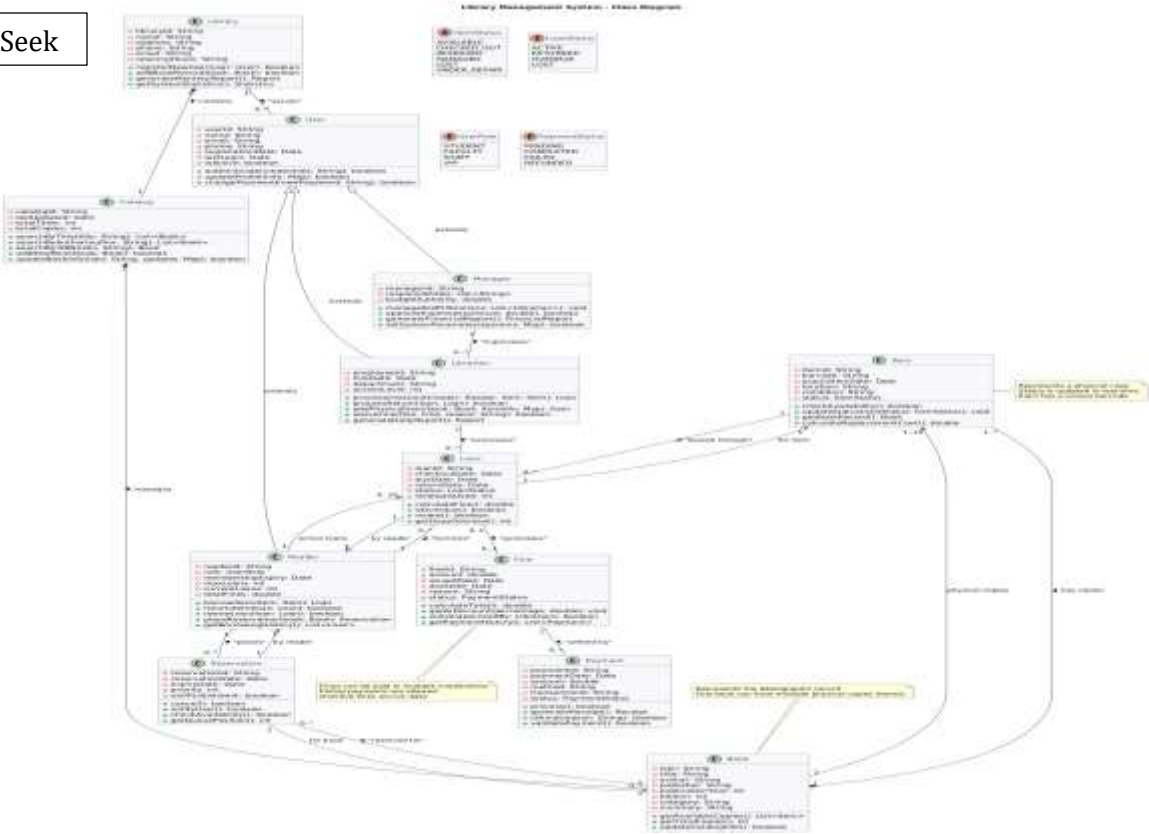
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> + waive(approvedBy: Librarian): boolean + getPaymentHistory(): List&lt;Payment&gt; }  class Payment { - paymentId: String - paymentDate: Date - amount: double - method: String - transactionId: String - status: PaymentStatus + process(): boolean + generateReceipt(): Receipt + refund(reason: String): boolean + validatePayment(): boolean }  ' ===== INHERITANCE RELATIONSHIPS ===== User &lt; -- Reader : extends User &lt; -- Librarian : extends User &lt; -- Manager : extends  ' ===== COMPOSITION RELATIONSHIPS ===== Library "1" *-- "1" Catalog : contains &gt; Catalog "1" *-- "0..*" Book : manages &gt; Book "1" *-- "1..*" Item : has copies &gt;  ' ===== ASSOCIATION RELATIONSHIPS ===== Reader "1" -- "0..*" Loan : "borrows" &gt; Item "1" -- "0..*" Loan : "loaned through" &gt; Loan "0..1" -- "0..1" Fine : "generates" &gt; Fine "1" -- "0..*" Payment : "settled by" &gt; Reader "1" -- "0..*" Reservation : "places" &gt; Book "1" -- "0..*" Reservation : "reserved for" &gt; Librarian "1" -- "0..*" Loan : "processes" &gt; Manager "1" -- "0..*" Librarian : "supervises" &gt;  ' ===== AGGREGATION RELATIONSHIPS ===== Library "1" o-- "0..*" User : "serves" &gt;  ' ===== MULTIPLICITIES ===== Reader "1" --&gt; "0..10" Loan : "active loans" Book "1" --&gt; "1..10" Item : "physical copies" Loan "1" --&gt; "1" Item : "for item" Loan "1" --&gt; "1" Reader : "by reader" Reservation "1" --&gt; "1" Book : "for book" Reservation "1" --&gt; "1" Reader : "by reader"  ' ===== NOTES ===== note top of Book Represents the bibliographic record One book can have multiple physical copies (Items) end note  note right of Item Represents a physical copy </pre> |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|  |                                                                                                                                                                                                                                                        |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>Status is updated in real-time<br/>Each has a unique barcode<br/>end note</p> <p>note bottom of Fine<br/>Fines can be paid in multiple installments<br/>Partial payments are allowed<br/>Overdue fines accrue daily<br/>end note</p> <p>@enduml</p> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

ChatGP



DeepSeek



Both diagrams satisfy Prompt 4 (inheritance, animations, properties + operations, relationship types, and multiplication) and render successfully (good). ChatGPT is more compact and readable, but it is less consistent in UML semantics (e.g., "union/collection" labels) and misses some obvious links (e.g., reader-reservation, good-reader). DeepSeek provides a more complete domain model with clear structure chains (library → catalog → book → item), rich constraints/notes, and strong coverage of transactions and payments. Overall, DeepSeek is a strong Prompt 4 solution due to its completeness and more consistent modeling.

## 6. Comparative Evaluation

Evaluate each combination (Prompt × Model × Diagram type) using the criteria below.

| Type     | Prompt | Model    | Code Renders | No Syntax errors | Overall(1-5) |
|----------|--------|----------|--------------|------------------|--------------|
| Use Case | 1      | ChatGPT  | OK           | No errors        | 5            |
| Use Case | 1      | DeepSeek | OK           | no errors        | 4            |
| Use Case | 2      | ChatGPT  | OK           | No errors        | 5            |
| Use Case | 2      | DeepSeek | OK           | Minor errors     | 3            |
| Use Case | 3      | ChatGPT  | OK           | No errors        | 4            |
| Use Case | 3      | DeepSeek | OK           | No errors        | 4            |
| Use Case | 4      | ChatGPT  | OK           | No errors        | 4            |
| Use Case | 4      | DeepSeek | OK           | No errors        | 4            |
| Class    | 1      | ChatGPT  | OK           | No errors        | 5            |
| Class    | 1      | DeepSeek | OK           | No errors        | 5            |
| Class    | 2      | ChatGPT  | OK           | No errors        | 4            |
| Class    | 2      | DeepSeek | OK           | No errors        | 4            |
| Class    | 3      | ChatGPT  | OK           | No errors        | 4            |
| Class    | 3      | DeepSeek | OK           | No errors        | 5            |
| Class    | 4      | ChatGPT  | OK           | No errors        | 4            |
| Class    | 4      | DeepSeek | OK           | No errors        | 5            |

## 7. Conclusions

- Best prompt (Use Case): ChatGPT Prompt1
- Best prompt (Class Diagram): DeepSeek Prompt 3

DeepSeek was superior because it provided a more accurate and rich classical structure for creating Use Case and Class Diagrams. In addition, DeepSeek demonstrated the correct Item / Book interactions with good use of strong relationships (composition, aggregation), which is very important for UML. ChatGPT provided simple, clear and standards-compliant procedures, but DeepSeek remained a step ahead with its extensive processing speed and detail. Although DeepSeek did not meet the Prompt 1 criteria, it was very good as a realistic model.